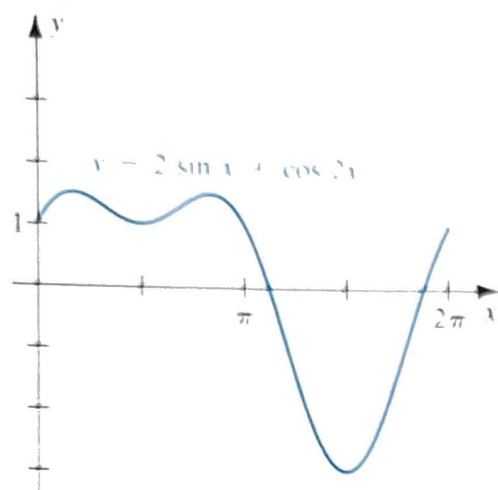


FIGURE 4.38



In Example 7 of Section 4.1 we found that the critical numbers of f in the interval $[0, 2\pi]$ are $\pi/6$, $5\pi/6$, $\pi/2$, and $3\pi/2$. Substituting these critical numbers for x in $f''(x)$, we obtain

$$f''(\pi/6) = 3, \quad f''(5\pi/6) = -3, \quad f''(\pi/2) = 2, \quad f''(3\pi/2) = 6.$$

Applying the second derivative test, we see that there are local maxima at $\pi/6$ and $5\pi/6$ and local minima at $\pi/2$ and $3\pi/2$. Thus, we have

$$\text{Local max: } f(\pi/6) = 3/2 \quad \text{and} \quad f(5\pi/6) = 3/2$$

$$\text{Local min: } f(\pi/2) = 1 \quad \text{and} \quad f(3\pi/2) = -3$$

Using this information and plotting several more points gives us the sketch in Figure 4.38.

EXERCISES 4.4

Exer. 1–18: Find the local extrema of f , using the second derivative test whenever applicable. Find the intervals on which the graph of f is concave upward or is concave downward, and find the x -coordinates of the points of inflection. Sketch the graph of f .

- | | |
|------------------------------------|--------------------------------|
| 1 $f(x) = x^3 - 2x^2 + x + 1$ | 4 $f(x) = 8x^2 - 2x^4$ |
| 2 $f(x) = x^3 + 10x^2 + 25x - 50$ | 6 $f(x) = 3x^5 - 5x^3$ |
| 3 $f(x) = 3x^4 - 4x^3 + 6$ | 8 $f(x) = x^4 - 4x^3 + 10$ |
| 5 $f(x) = 2x^6 - 6x^4$ | 10 $f(x) = 2 - \sqrt[3]{x^2}$ |
| 7 $f(x) = (x^2 - 1)^2$ | 12 $f(x) = x^{2/3}(1 - x)$ |
| 9 $f(x) = \sqrt[3]{x} - 1$ | 14 $f(x) = x\sqrt[3]{3x + 2}$ |
| 11 $f(x) = \sqrt[3]{x^2}(3x + 10)$ | 16 $f(x) = 6x^{1/2} + x^{3/2}$ |
| 13 $f(x) = x^2(3x - 5)^{1/3}$ | 18 $f(x) = x\sqrt{4 - x^2}$ |
| 15 $f(x) = 8x^{1/3} + x^{4/3}$ | |
| 17 $f(x) = x^2\sqrt{9 - x^2}$ | |

Exer. 19–24: Use the second derivative test to find the local extrema of f on the interval $[0, 2\pi]$. (These exercises are the same as Exercises 17–22 in Section 4.3, for which the method of solution involved the first derivative test.)

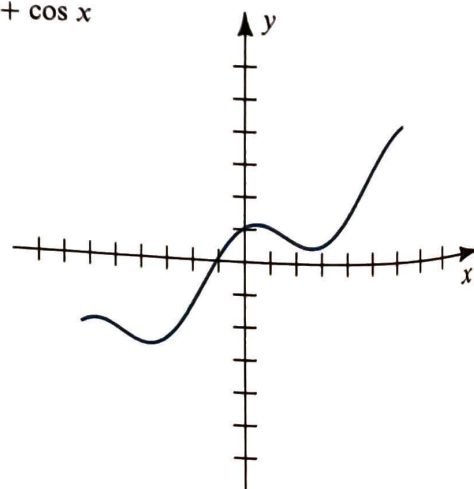
- | | |
|-----------------------------------|--------------------------------|
| 19 $f(x) = \cos x + \sin x$ | 20 $f(x) = \cos x - \sin x$ |
| 21 $f(x) = \frac{1}{2}x - \sin x$ | 22 $f(x) = x + 2 \cos x$ |
| 23 $f(x) = 2 \cos x + \sin 2x$ | 24 $f(x) = 2 \cos x + \cos 2x$ |

Exer. 25–28: Use the second derivative test to find the local extrema of f on the given interval. (See Exercises 29–32 of Section 4.3.)

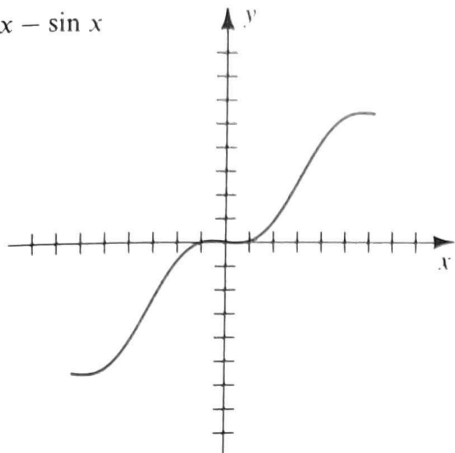
- | | |
|----------------------------------|-------------------|
| 25 $f(x) = \sec \frac{1}{2}x;$ | $[-\pi/2, \pi/2]$ |
| 26 $f(x) = \cot^2 x + 2 \cot x;$ | $[\pi/6, 5\pi/6]$ |
| 27 $f(x) = 2 \tan x - \tan^2 x;$ | $[-\pi/3, \pi/3]$ |
| 28 $f(x) = \tan x - 2 \sec x;$ | $[-\pi/4, \pi/4]$ |

Exer. 29–30: Shown in the figure is a graph of the equation for $-2\pi \leq x \leq 2\pi$. (See Exercises 33–34 of Section 4.3.) Use the second derivative test to find the local extrema of f .

$$29 \quad y = \frac{1}{2}x + \cos x$$



30 $y = \frac{\sqrt{3}}{2}x - \sin x$



Exer. 31–38: Sketch the graph of a continuous function f that satisfies all of the stated conditions.

- 31 $f(0) = 1$; $f(2) = 3$; $f'(0) = f'(2) = 0$;
 $f'(x) < 0$ if $|x - 1| > 1$; $f'(x) > 0$ if $|x - 1| < 1$;
 $f''(x) > 0$ if $x < 1$; $f''(x) < 0$ if $x > 1$
- 32 $f(0) = 4$; $f(2) = 2$; $f(5) = 6$; $f'(0) = f'(2) = 0$;
 $f'(x) > 0$ if $|x - 1| > 1$; $f'(x) < 0$ if $|x - 1| < 1$;
 $f''(x) < 0$ if $x < 1$ or if $|x - 4| < 1$;
 $f''(x) > 0$ if $|x - 2| < 1$ or if $x > 5$
- 33 $f(0) = 2$; $f(2) = f(-2) = 1$; $f'(0) = 0$;
 $f'(x) > 0$ if $x < 0$; $f'(x) < 0$ if $x > 0$;
 $f''(x) < 0$ if $|x| < 2$; $f''(x) > 0$ if $|x| > 2$
- 34 $f(1) = 4$; $f'(x) > 0$ if $x < 1$; $f'(x) < 0$ if $x > 1$;
 $f''(x) > 0$ for every $x \neq 1$
- 35 $f(-2) = f(6) = -2$; $f(0) = f(4) = 0$; $f(2) = f(8) = 3$;
 f' is undefined at 2 and 6; $f'(0) = 1$;
 $f'(x) > 0$ throughout $(-\infty, 2)$ and $(6, \infty)$;
 $f'(x) < 0$ if $|x - 4| < 2$;
 $f''(x) < 0$ throughout $(-\infty, 0)$, $(4, 6)$, and $(6, \infty)$;
 $f''(x) > 0$ throughout $(0, 2)$ and $(2, 4)$
- 36 $f(0) = 2$; $f(2) = 1$; $f(4) = f(10) = 0$; $f(6) = -4$;
 $f'(2) = f'(6) = 0$;

$f'(x) < 0$ throughout $(-\infty, 2)$, $(2, 4)$, $(4, 6)$, and $(10, \infty)$;
 $f'(x) > 0$ throughout $(6, 10)$;
 $f'(4)$ and $f'(10)$ do not exist;
 $f''(x) > 0$ throughout $(-\infty, 2)$, $(4, 10)$, and $(10, \infty)$;
 $f''(x) < 0$ throughout $(2, 4)$

- 37 If n is an odd integer, then $f(n) = 1$ and $f'(n) = 0$; if n is an even integer, then $f(n) = 0$ and $f'(n)$ does not exist; if n is any integer, then

- (a) $f'(x) > 0$ whenever $2n < x < 2n + 1$
 (b) $f'(x) < 0$ whenever $2n - 1 < x < 2n$
 (c) $f''(x) < 0$ whenever $2n < x < 2n + 2$

- 38 $f(x) = x$ if $x = -1, 2, 4$, or 8 ;
 $f'(x) = 0$ if $x = -1, 4, 6$, or 8 ;
 $f'(x) < 0$ throughout $(-\infty, -1)$, $(4, 6)$, and $(8, \infty)$;
 $f'(x) > 0$ throughout $(-1, 4)$ and $(6, 8)$;
 $f''(x) > 0$ throughout $(-\infty, 0)$, $(2, 3)$, and $(5, 7)$;
 $f''(x) < 0$ throughout $(0, 2)$, $(3, 5)$, and $(7, \infty)$

- 39 Prove that the graph of a quadratic function has no point of inflection. State conditions for which the graph is always

- (a) concave upward (b) concave downward

- 40 Prove that the graph of a polynomial function of degree 3 has exactly one point of inflection.

- [c] Exer. 41–42:** Graph f on $[-1, 1]$. (a) Estimate where the graph of f is concave upward or is concave downward. (b) Estimate the x -coordinate of each point of inflection.

41 $f(x) = 4x^5 + x^4 + 3x^2 - 2x + 1$

42 $f(x) = (x - 0.1)^2 \sqrt{1.08 - 0.9x^2}$

- [c] Exer. 43–44:** Graph f'' on $[0, 3]$. (a) Estimate where the graph of f is concave upward or is concave downward. (b) Estimate the x -coordinate of each point of inflection.

43 $f''(x) = x^4 - 5x^3 + 7.57x^2 - 3.3x + 0.4356$

44 $f''(x) = 2.1 \sin \pi x + 1.4 \cos x - 0.6$

4.5 SUMMARY OF GRAPHICAL METHODS

Many applications of calculus are based on the behavior of the function values $f(x)$ of a function f as x varies throughout a set of real numbers. One of the best ways to exhibit this behavior is to sketch the graph of f , because a graph displays properties of f that may otherwise be hidden or unclear. Up to this point in the text we have discussed a variety of topics that are useful for sketching graphs. These discussions have been scattered throughout the first four chapters, so we shall summarize this work here and consider several additional examples.